ENES 220 Exam #2: Solving Methodology for an AI Model

0. Operating Rules (read first, obey always)

1. **Allowed tools only.** Use ONLY:
   - The formulas in Section 1 (copied from the official Exam #2 equation sheet).
   - Static equilibrium ($\Sigma F = 0$, $\Sigma M = 0$) to find reactions, internal V and M.
   - The shape table in Section 1.3 (rectangle, triangle, semicircle, circle).
   - Unit conversions.
   - Algebraic rearrangement of the above.
2. **Do NOT use** shortcut formulas that are not on the sheet. Examples of forbidden shortcuts: $\tau_{max} = 3V/2A$, $\tau = 1.5V/A$, $\sigma = 6M/bh^2$, $S = I/c$ section modulus tables, $wL^2/8$ or PL quoted from memory (these must be derived from equilibrium, see Section 2.7).
3. **Material properties** (E, allowables, fastener strength) come ONLY from the problem statement. If a value is missing, stop and ask for it. Never assume a textbook value.
4. **Show every step**: piece table, centroid, I, Q, final substitution with units.
5. **Units**: pick one consistent system before computing.
   - SI: N, mm, MPa (1 MPa = 1 N/mm^2). Convert kN·m to N·mm by $\times 10^6$. Convert kN to N by $\times 10^3$. Convert GPa to MPa by $\times 10^3$.
   - US: lb, in, psi, or kip, in, ksi. Convert ft to in by $\times 12$. Convert kip·ft to kip·in by $\times 12$.
   - kPa = 10^{-3} MPa. If an allowable is in kPa, convert to MPa ($\div 1000$) before using N and mm.
6. **"Check all conditions"** means compute the limit from EVERY possible failure mode and report the SMALLEST as governing. List each one.
7. **"Magnitude and sense"** means report $|\sigma|$ and state tension (T) or compression (C).

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1. Allowed Formulas (verbatim from the equation sheet)

1.1 Stresses in Beams

Formula	Meaning
$\sigma_x = -My / I_z$	Bending stress at distance y from the NA (y positive upward)
$\sigma_{max} = Mc / I_z$, $c = y_{max}$	Max bending stress magnitude, c = farthest fiber distance from NA
$\sigma_x = E\,\varepsilon_x$, $\varepsilon_x = -y/\rho$, $M = E\,I_z / \rho$	Strain, curvature relations
$\tau = V\,Q_i / (I_z\,b_i)$	Transverse shear stress at a cut
$Q_i = \int y\,dA = \Sigma\,\bar{y}_i\,A_i$	First moment of the area on ONE side of the cut, about the NA
$q = \tau_jt\,b_jt = V\,Q_jt / I_z$	Shear flow at a joint (force per length)
$s_{max} = F_s / q = n\,F_s\,I_z / (V\,Q_jt)$	Max fastener spacing, n = fasteners per row crossing that joint
$k = E_1 / E_2$	Transformation factor (material 1 transformed into material 2)
$b_1T = k\,b_1$	Transformed width of material 1 (height unchanged)
$\sigma_1 = -k\,M\,y_1 / I_Tz$	Stress in material 1 (the transformed one)
$\sigma_2 = -M\,y_2 / I_Tz$	Stress in material 2 (the reference one)

1.2 Centroids and Moments of Inertia

- $\bar{y}_c = \Sigma\,\bar{y}_i\,A_i / \Sigma\,A_i$
- $I_z = \Sigma\,(I_zc_i + A_i\,d_i^2)$, with $d_i = \bar{y}_i - \bar{y}_c$

1.3 Shape Table

Shape	Area	Centroid $\bar{y}$ (from base)	I about own centroid
Rectangle	bh	$h/2$	$bh^3/12$ (about base: $bh^3/3$)
Triangle	$bh/2$	$h/3$	$bh^3/36$ (about base: $bh^3/12$)
Semicircle	$\pi r^2/2$	$4r/(3\pi)$	$\pi r^4/8 - 8r^4/(9\pi)$ (about diameter: $\pi r^4/8$)
Circle	πr^2	r	$\pi r^4/4$ (about tangent base: $5\pi r^4/4$)

2. Methodologies by Problem Type

2.1 Core Procedure: Neutral Axis and I_z (used by almost every problem)

1. Choose a reference line y = 0 (usually the bottom edge).
2. Split the cross section into simple shapes from Section 1.3. Hollow regions may be handled as negative area.
3. Build a table: piece | b | h | A_i | $\bar{y}_i$ | $A_i\bar{y}_i$ | I_{zc_i} | d_i | $A_i d_i^2$.
4. $\bar{y}_c = \Sigma A_i\bar{y}_i / \Sigma A_i$. If the section is symmetric top to bottom, $\bar{y}_c$ is mid height (state this).
5. $I_z = \Sigma(I_{zc_i} + A_i d_i^2)$. For negative (hole) pieces, subtract both terms.
6. Find c_top = (total height − $\bar{y}_c$) and c_bot = $\bar{y}_c$.
7. If the problem GIVES $\bar{y}$ or I_z, use the given values (do not recompute unless asked).

2.2 Max Bending Stress (homogeneous beam)

1. Get M (given, or from Section 2.7).
2. Run Section 2.1.
3. $\sigma_{top} = -M(c_{top})/I_z$, $\sigma_{bot} = -M(-c_{bot})/I_z = +M\, c_{bot}/I_z$.
   - Positive M: top in compression, bottom in tension.
   - Negative M: top in tension, bottom in compression.
4. σ_{max} magnitude = M c / I_z with c = larger of c_top, c_bot. State location (top or bottom fiber) and T or C.
5. **If the M diagram has both positive and negative values** (e.g. M goes from −4 to +6 kN·m) and the section is unsymmetric, compute stress at BOTH extreme fibers for BOTH the max positive and max negative M. Report the overall largest. A smaller |M| with a larger c can govern.

2.3 Max Allowable Moment (homogeneous, given σ_allowable)

1. Section 2.1.
2. $M_{allow} = \sigma_{allow} I_z / c$ using the larger c (if one allowable for both T and C).
3. If tension and compression allowables differ: compute M from each fiber with its own allowable based on whether that fiber is in T or C, and take the smaller.

2.4 Composite Beams (Transformed Section)

1. Identify material 1 (the one being transformed) and material 2 (the reference). State the choice. Convention: $k = E_1/E_2$.
   - Common choice: transform the stiffer material into the softer one (k > 1), but either works.
2. $b_1T = k\, b_1$ for every material 1 piece. Heights stay the same. Material 2 widths stay actual.

3. Run Section 2.1 on the transformed section to get $\bar{y}_c$ and I_Tz.
4. Stresses:
   - Material 2 (reference): $\sigma_2 = -M\, y_2 / I_Tz$
   - Material 1 (transformed): $\sigma_1 = -k\, M\, y_1 / I_Tz$
   - y measured from the NA to the extreme fiber OF THAT MATERIAL (not necessarily the extreme of the whole section).
5. Report magnitude and sense (T or C) for each material using the sign of −My.
6. **Max allowable moment (composite):** for each material, find its farthest fiber from the NA, then
   - Material 2: $M = \sigma_allow,2\, I_Tz / |y_2|$
   - Material 1: $M = \sigma_allow,1\, I_Tz / (k\,|y_1|)$
   - Check every material (and every fiber if allowables differ for T vs C). Smallest M governs.
7. With 3+ materials, pick one reference; each other material gets its own k = E_i/E_ref.

2.5 Transverse Shear Stress at a Cut

1. Section 2.1 (I_z and $\bar{y}_c$).
2. Choose the cut (horizontal line at the point of interest).
3. $Q = \Sigma\, A_i\, (\bar{y}_i - \bar{y}_c)$ for the area on ONE side of the cut (use whichever side is simpler; |Q| is the same). $\bar{y}_i - \bar{y}_c$ is the distance from the NA to each piece's own centroid.
4. b = total material width actually cut by the line (sum all webs/legs crossed).
5. $\tau = V\, Q / (I_z\, b)$.
6. **Max shear stress** occurs at the NA for standard sections: Q = first moment of everything above (or below) the NA. If the width is much narrower somewhere else, also check that cut and take the larger τ.
7. Composite beams: use transformed areas in Q and I_Tz, then divide by the ACTUAL width b at the cut ($q = VQ/I_Tz$, $\tau = q/b$).

2.6 Joints: Glue, Nails, Screws, Bolts, Welds

1. Identify the piece that would slide off at the joint (the "attached" piece).
2. $Q_jt = A_piece \times (\bar{y}_piece - \bar{y}_c)$ using ONLY that attached piece (or everything beyond the joint).
3. Shear flow $q = V\, Q_jt / I_z$.
4. Glue or weld stress: $\tau_jt = q / b_jt$, where b_jt = total glued/welded contact width across that joint line (sum over all bond lines carrying that piece).

5. Fastener spacing: $s_{max} = n F_s I_z / (V Q_{jt})$, n = number of fasteners in one row resisting that joint (count fasteners at the same longitudinal position along that joint).
6. Max V from fasteners at known spacing: $V = n F_s I_z / (s Q_{jt})$.
7. Max V from wood/material shear: $V = \tau_{allow} I_z b / Q$, checked at the NA and at each joint.
8. Unsymmetric built-up sections can have different joints with different Q, s, n, orientation. Check EACH joint separately.

2.7 Finding Vmax and Mmax from a Loading (use equilibrium, not memorized results)

1. Draw the FBD and solve reactions with $\Sigma F_y = 0$ and $\Sigma M = 0$.
2. Cut the beam at a general x, write V(x) and M(x) from equilibrium of one side.
3. Vmax = largest |V(x)|. Mmax = largest |M(x)|; for distributed loads it occurs where V = 0 (set V(x) = 0, solve x, substitute).
4. Keep signs: positive M = sagging (compression on top). The sign matters for unsymmetric sections.
5. If V and M diagrams are given, read Vmax, M_{max+}, M_{max-} directly.

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3. Common Pitfalls Checklist

- ☐ c is measured from the NA, not from the mid height, unless the section is symmetric.
- ☐ b in $\tau = VQ/(Ib)$ is the width AT THE CUT, summed across all pieces cut.
- ☐ Q uses only the area on one side of the cut; distances are to the NA.
- ☐ For joints, Q is the attached piece only.
- ☐ n counts fasteners per row for that joint.
- ☐ In composite beams, the transformed material's stress includes the extra factor k.
- ☐ In composite beams, each material's max stress is at ITS OWN farthest fiber.
- ☐ Unit consistency: N with mm gives MPa; lb with in gives psi; kip with in gives ksi.
- ☐ M given in kN·m: multiply by 10^6 for N·mm. M given in kip·ft: multiply by 12 for kip·in.
- ☐ "Check all conditions": list every limit and choose the smallest.
- ☐ When the problem gives $\bar{y}$ or I_z, use them.

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4. Worked Reference Examples (verify procedure)

Example A: Composite bending (wood with steel straps top and bottom)

Given: wood 200 × 300 mm, E_w = 10 GPa; two steel straps 200 × 20 mm, E_s = 200 GPa; M = 5 kN·m.

1. Material 1 = steel, material 2 = wood. $k = E_1/E_2 = 200/10 = 20$.
2. $b_1T = 20(200) = 4000$ mm.

3. Symmetric, so NA at mid height (170 mm from bottom).
4. I_Tz:
   - Wood: $200(300)^3/12 = 450.0 \times 10^6$ mm^4
   - Each strap: $4000(20)^3/12 + (4000 \times 20)(160)^2 = 2.667 \times 10^6 + 2048 \times 10^6 = 2050.7 \times 10^6$ mm^4
   - $I_{Tz} = 450.0 \times 10^6 + 2(2050.7 \times 10^6) = 4551.3 \times 10^6$ mm^4
5. $M = 5 \times 10^6$ N·mm.
   - Wood ($y_2 = 150$ mm): $|\sigma_2| = (5 \times 10^6)(150)/(4551.3 \times 10^6) = 0.165$ MPa
   - Steel ($y_1 = 170$ mm): $|\sigma_1| = 20(5 \times 10^6)(170)/(4551.3 \times 10^6) = 3.74$ MPa

Example B: Fastener spacing (box beam, four boards, screws)

Given: outer 6 in wide × 8 in tall; top and bottom boards 6 × 1 in; side boards 1 × 6 in between them; F = 210 lb per screw; V = 1150 lb; screws pass vertically through the top board into the two side boards.

1. Symmetric, NA at 4 in.
2. $I_z = 6(8)^3/12 - 4(6)^3/12 = 256 - 72 = 184$ in^4 (outer rectangle minus hollow).
3. Attached piece = top board: $Q_{jt} = (6 \times 1)(3.5) = 21$ in^3.
4. n = 2 (one screw per side in each row).
5. $s_{max} = n F I_z / (V Q_{jt}) = 2(210)(184)/(1150 \times 21) = 3.20$ in.

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5. Required Output Format for Each Answer

1. Given data with units, and unit system chosen.
2. Material assignment (composite) and k.
3. Piece table, $\bar{y}_c$, I_z (or I_{Tz}).
4. For each requested quantity: formula from Section 1, substitution, result with units.
5. For "check all conditions": list each limit, then state the governing value and why.
6. Final answers boxed or bolded, with location and T/C where relevant.